

Multi-Modal Fusion Techniques for Detecting Abnormal Events in Videos

Presented by AIP491_G9

Team member

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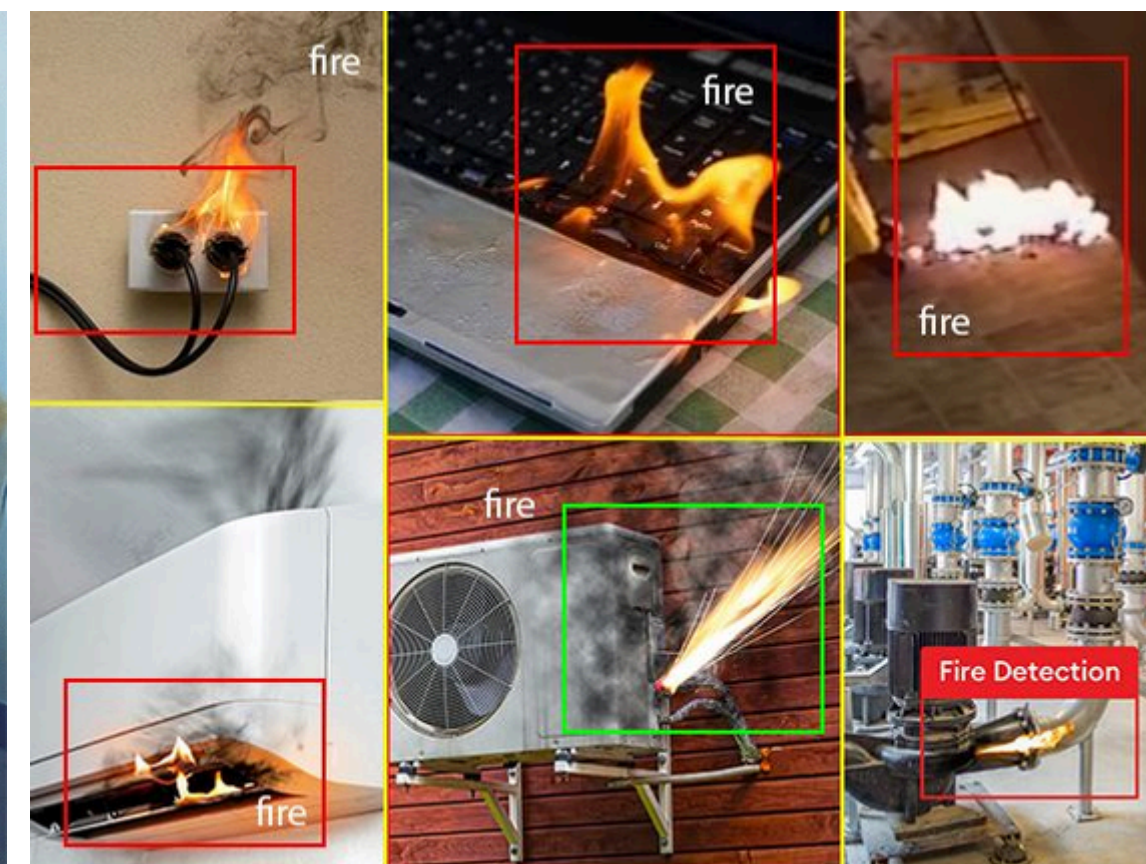
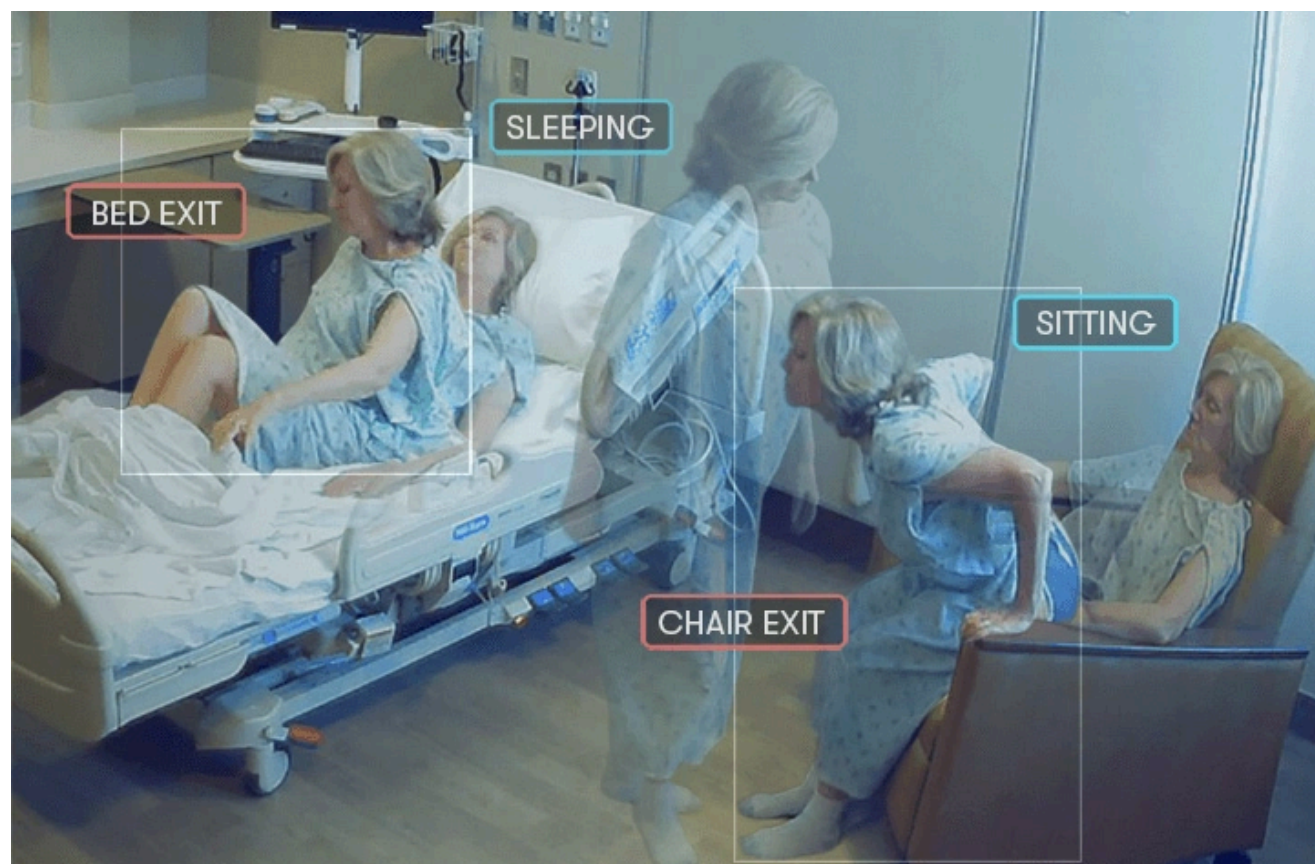
Abnormal Event

Abnormal events in videos refer to rare and context-dependent behaviors that deviate significantly from normal patterns



Why Detect Abnormal Events?

- **Security & Traffic:** Early detection of dangerous behaviors, accidents, ...
- **Health Monitoring:** Spot abnormal signs early.
- **Fire & Emergency:** Catch fire risks and dangerous acts fast.



Current Approaches

1. Handcrafted Methods

- Optical Flow, Trajectories
- Traditional ML

2. Deep Learning

- CNNs: Spatial features
- RNNs (LSTM): Temporal modeling

3. Transformers

- Self-attention for global context

4. Self-Supervised Learning

- Tasks: Temporal order, contrastive

5. Hybrid Models

- Combine models/modalities
-

Limitations

- RGB-sensitive to noise & lighting
 - Fusion strategies lack adaptability
 - Cross-Modal Misalignment
-

Objectives

- Add Pose Modality
- Adaptive Fusion with Gated Attention
- Cross-Modal Alignment via Contrastive Loss

Related Work

Yue Wang et al. (2023) – M3DM

- 3D Point Cloud + RGB Fusion
- Contrastive Learning + Memory Bank

Ghadiya et al. (2024) – Cross-Modal Fusion

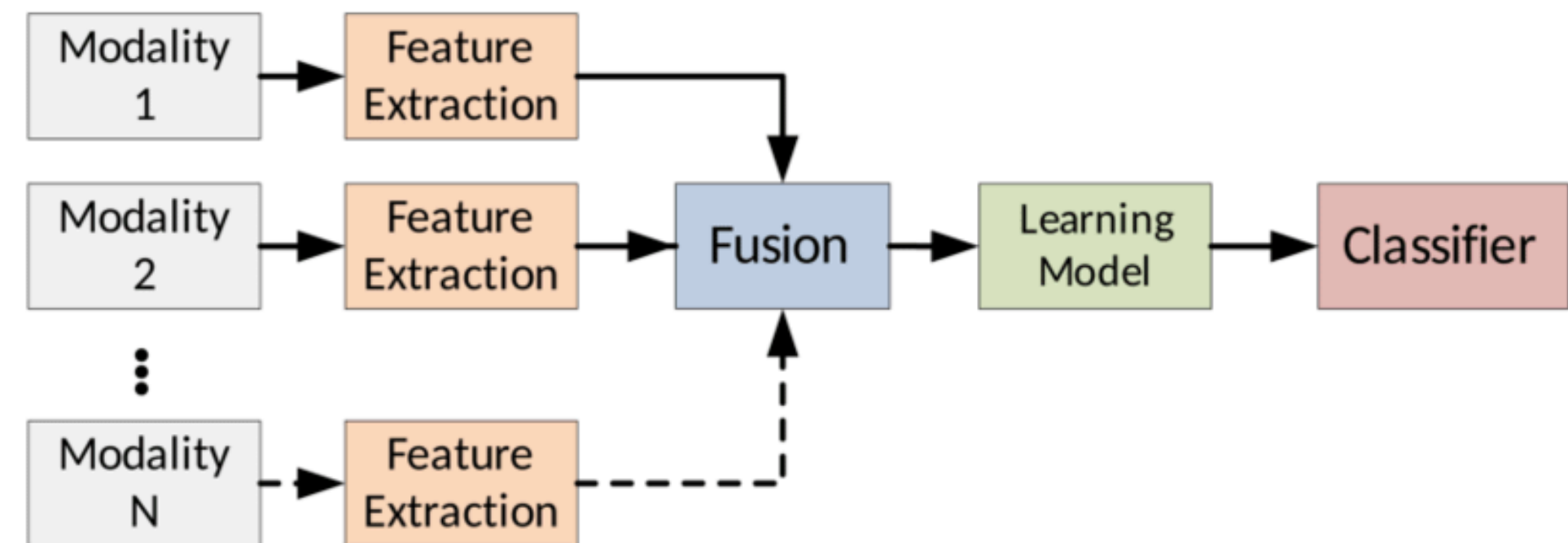
- Audio-Visual Feature Selection
- Lorentzian Graph Attention

Wenwen Sun et al. (2024) – Multimodal Multiscale Fusion

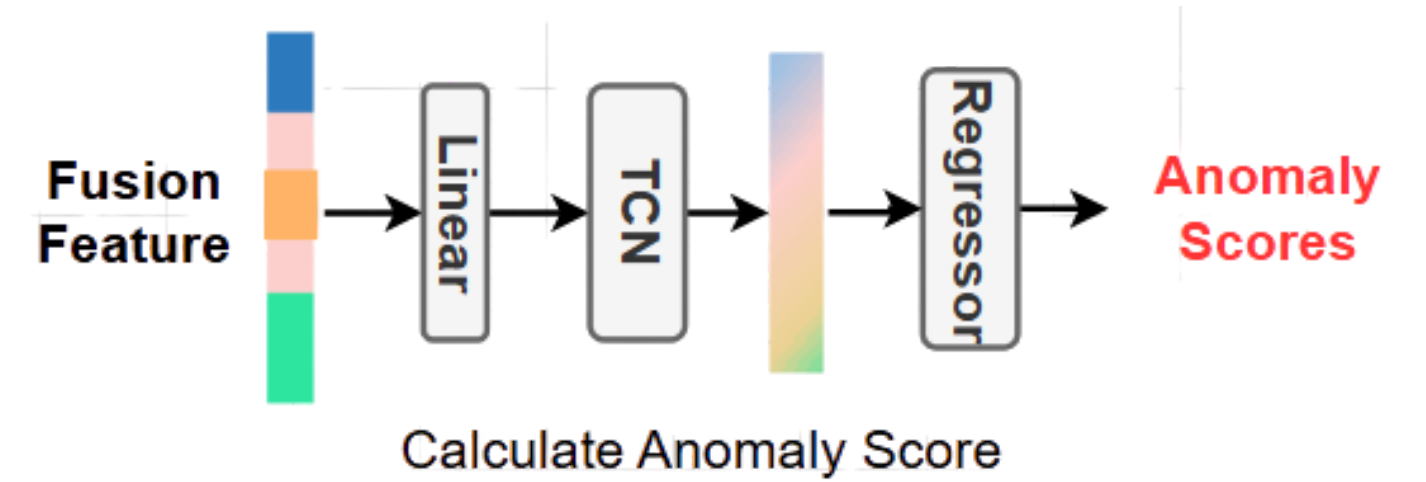
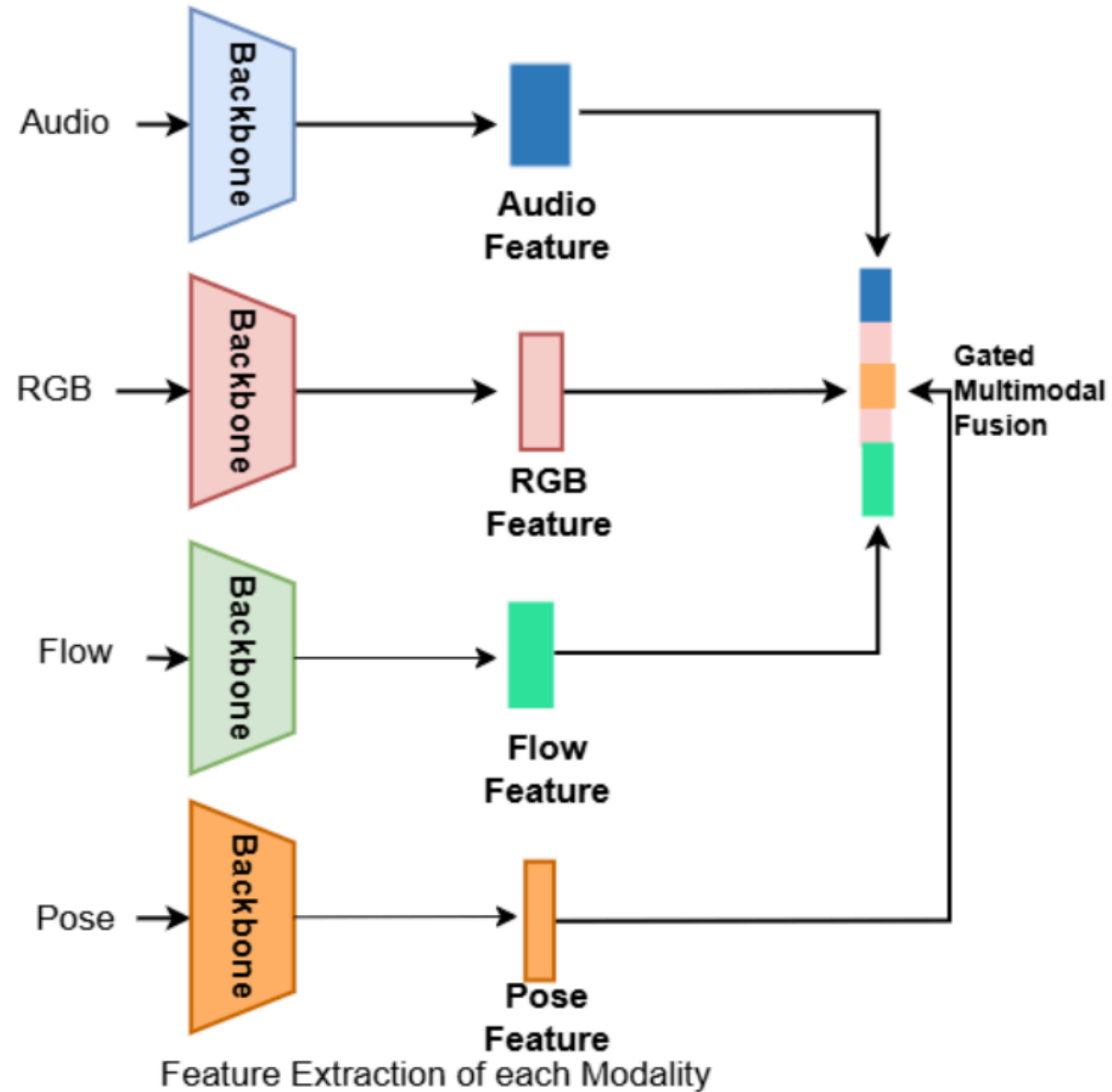
- Multi-Level Feature Aggregation
- Attention-based Modality Balance

Wenping Jin et al. (2024) – Aligning First, Then Fusing

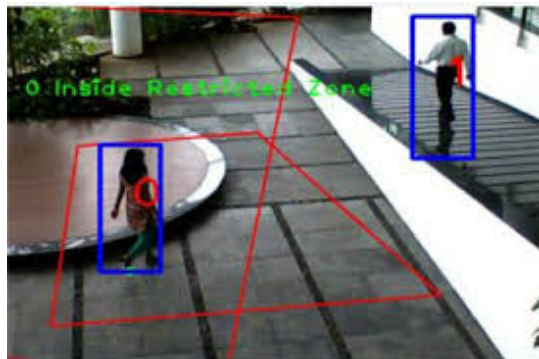
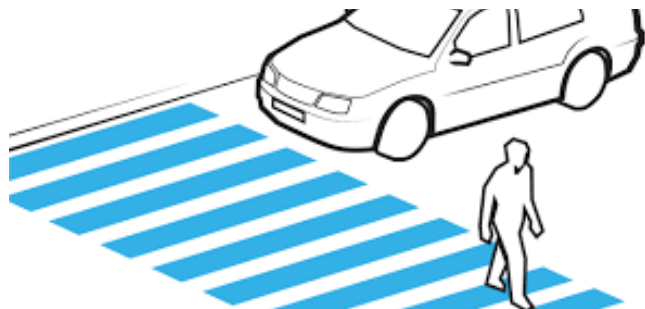
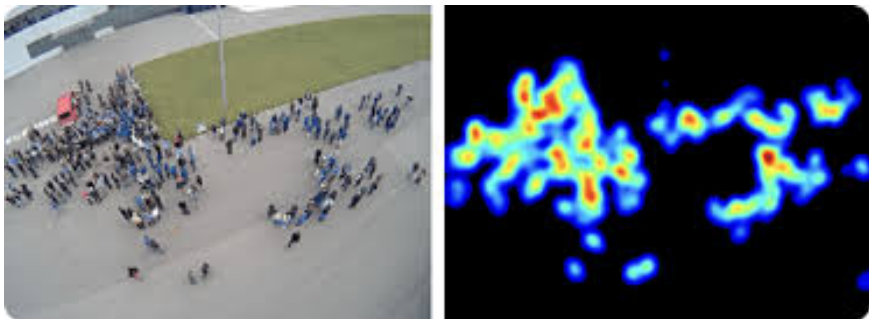
- Multimodal Semantic Feature Alignment
- MFMS (Modality-wise Feature Matching Subspace)



Methodology



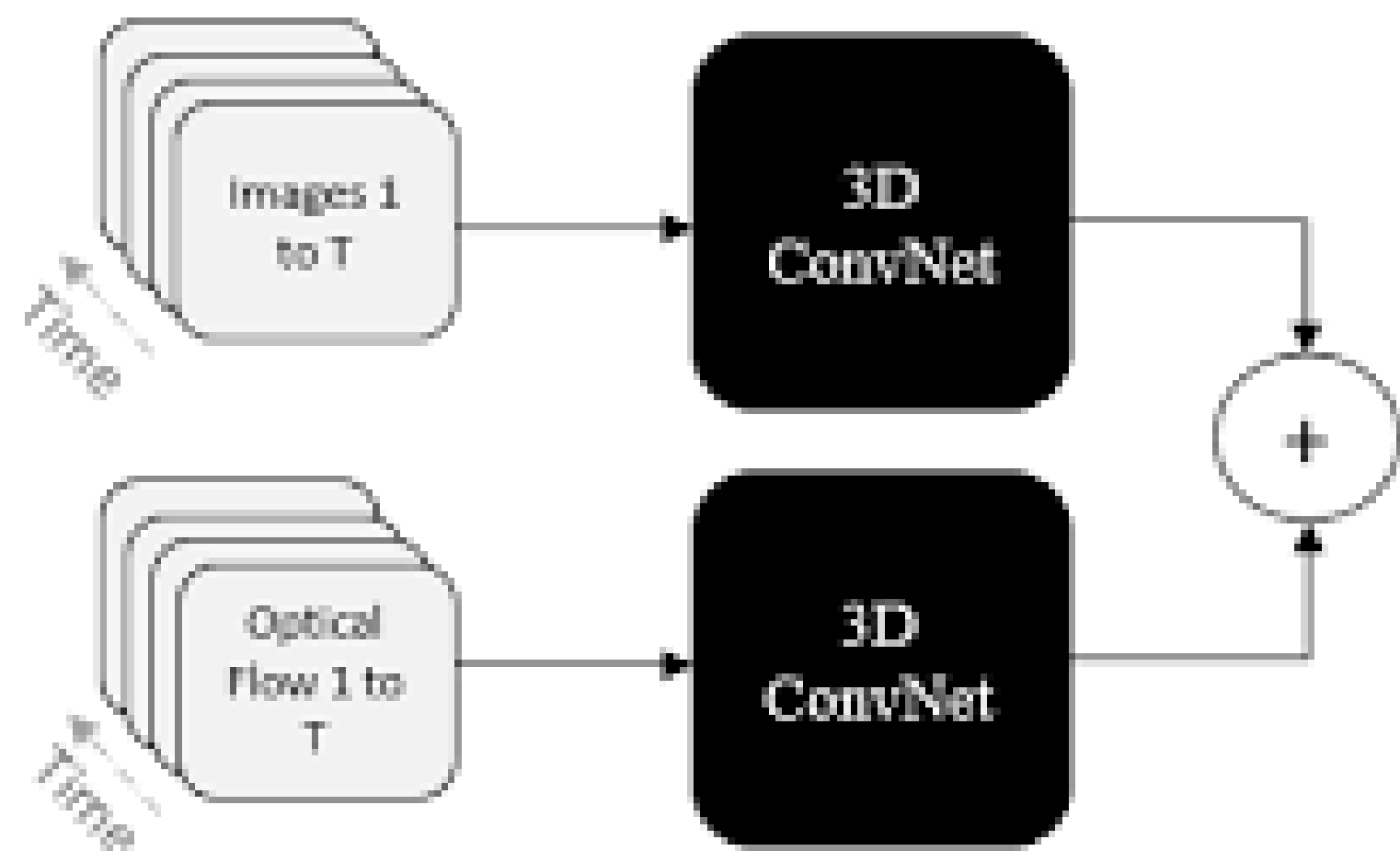
RGB and Flow Feature Extraction

Abnormal situation	Detected through	Reason
	Optical Flow	Abnormal fast movement
	RGB + Optical Flow	Out-of-context object + Irregular motion
	Optical Flow	Signs of evasive behavior – possibly due to an incident

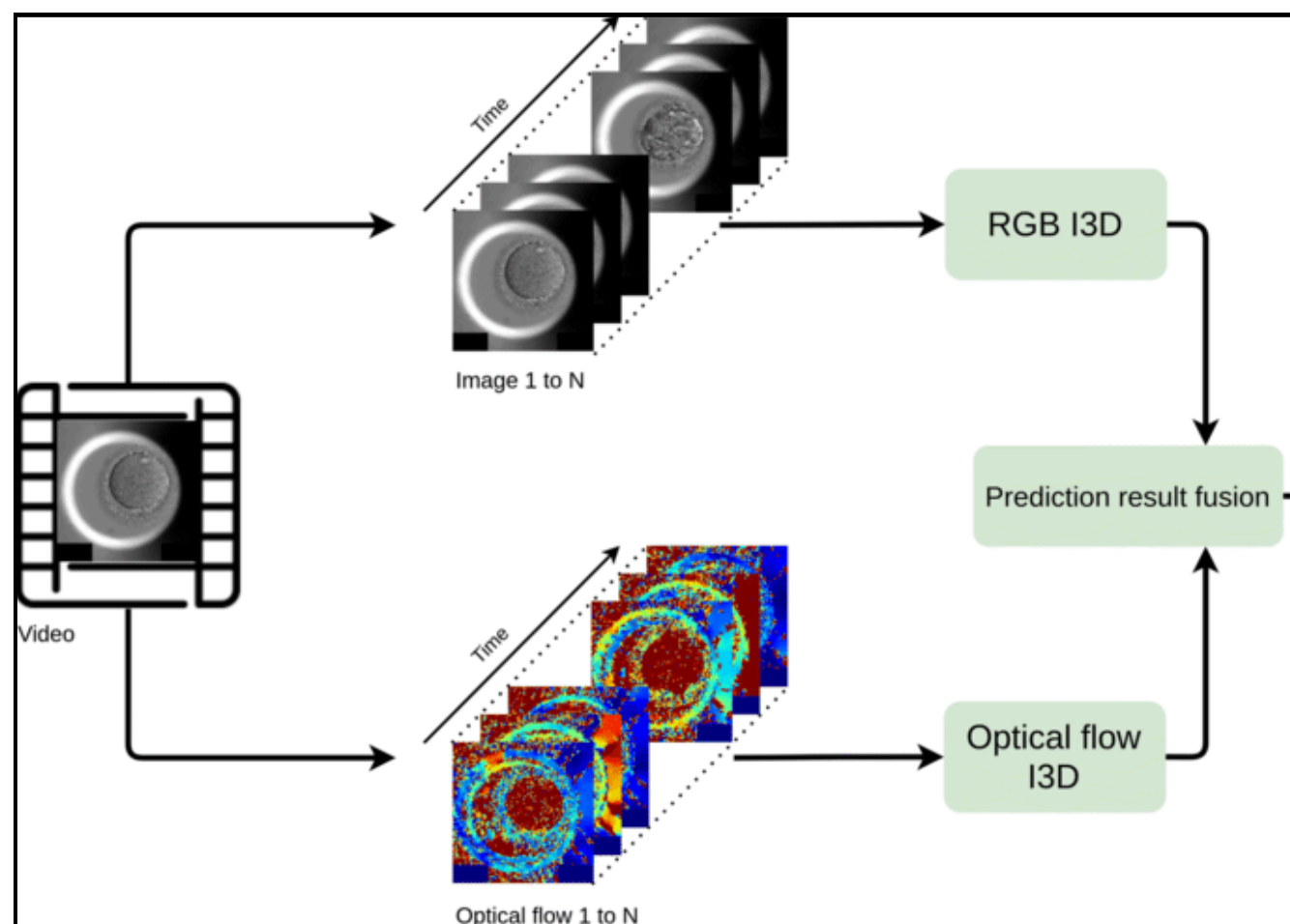
RGB and Flow Feature Extraction

I3D model:

- Two-Stream Support
- Pretrained on Kinetics-400
- Good results when combined with VGG model on xd-violence dataset



RGB and Flow Feature Extraction

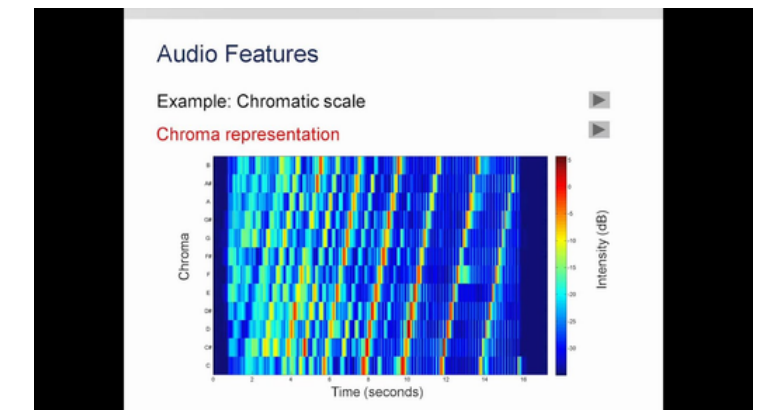


Input RGB: $X_R \in \mathbb{R}^{T \times H \times W \times 3}$

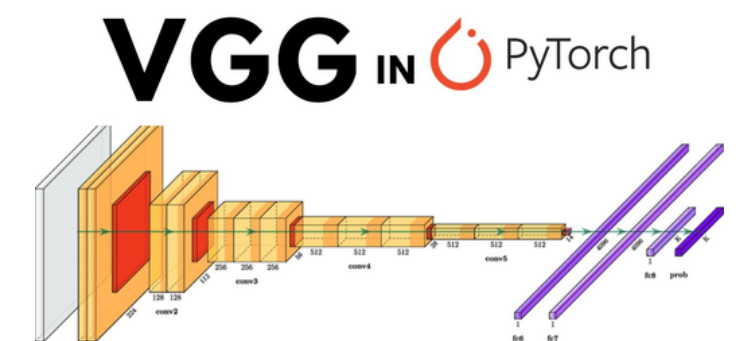
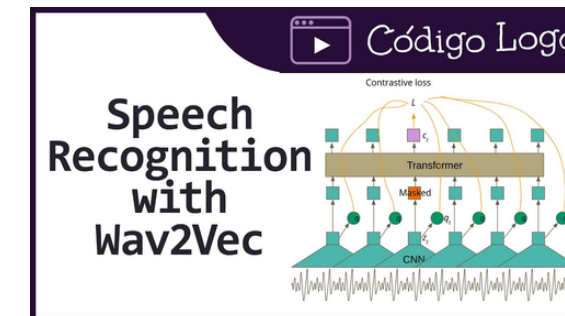
Input Flow: $X_F \in \mathbb{R}^{T \times H \times W \times 2}$

Audio feature extraction

Traditional ways (MFCC,..)

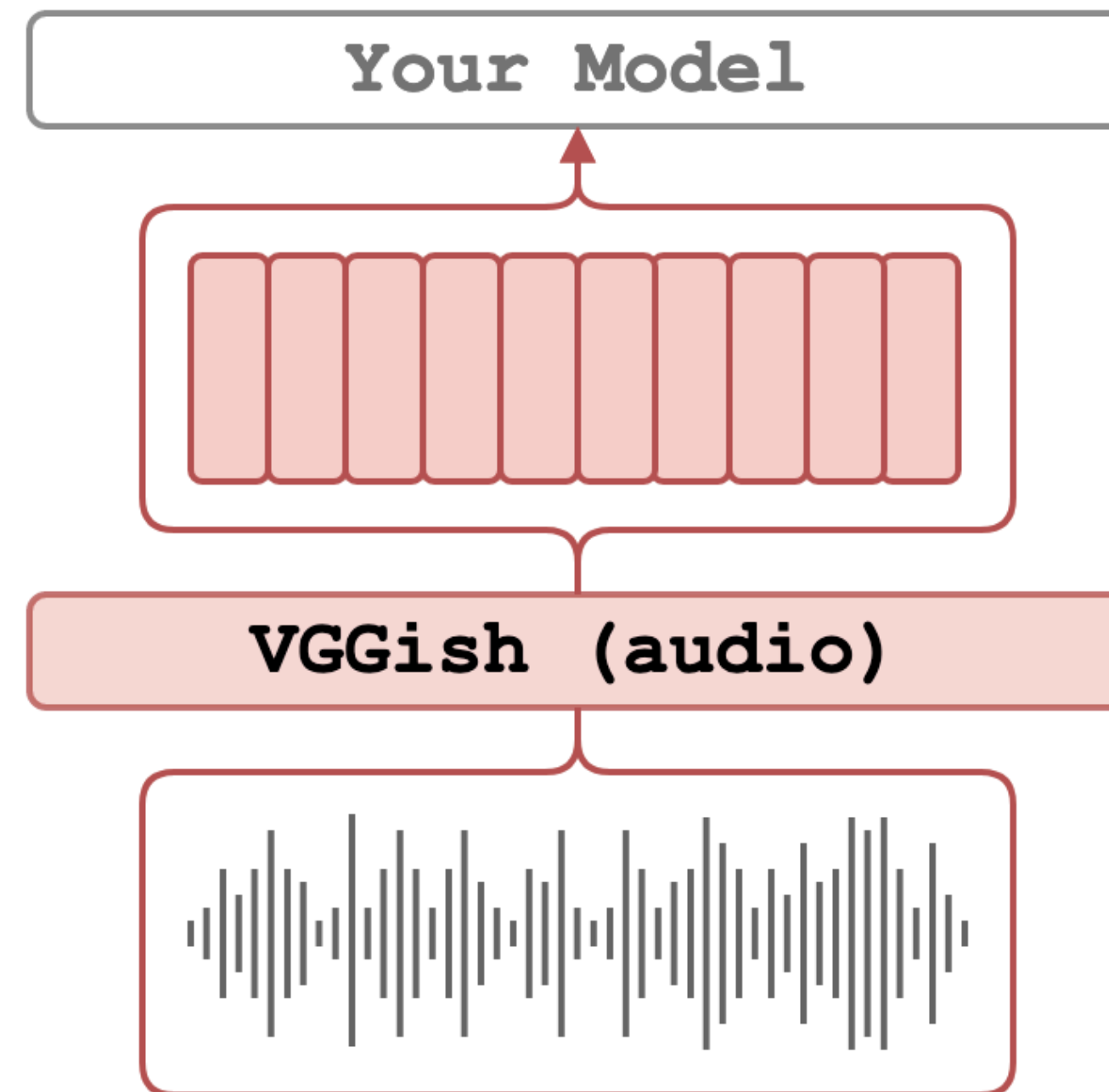


Deep-learning (VGG,
YAMNeT,Wav2Vec,...)

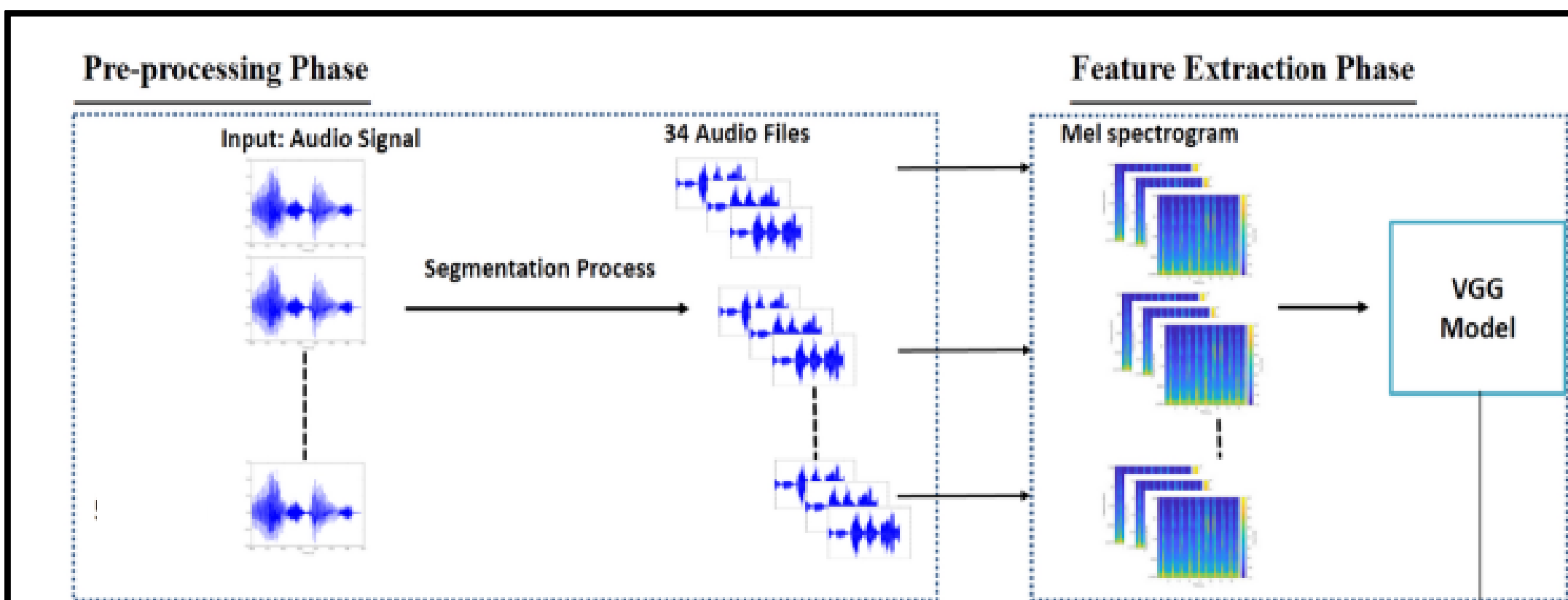


Audio feature extraction

- Pretrained on YouTube-100M.
- Good results when combined with I3D model on xd-violence dataset



Audio feature extraction

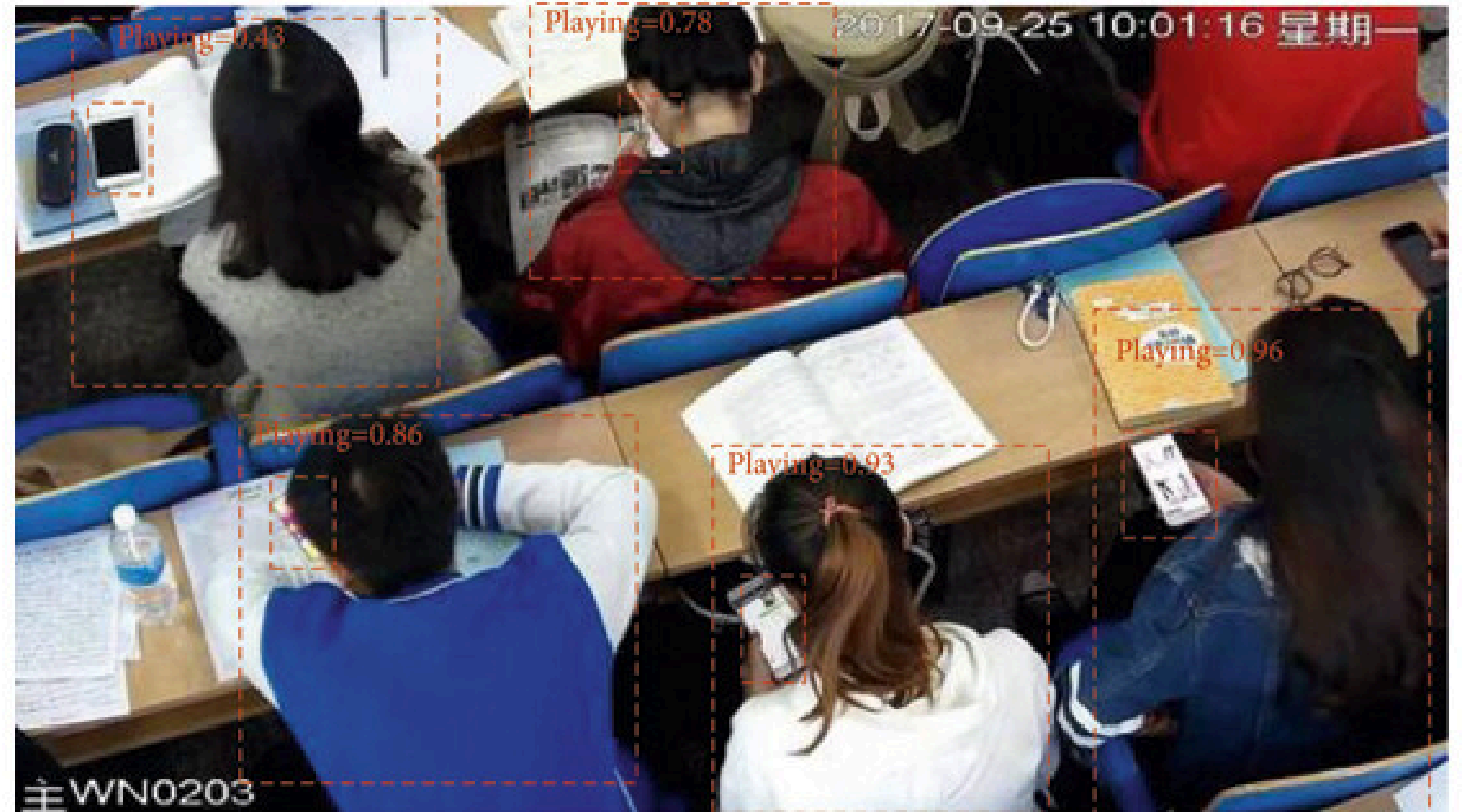


- **Input:** 0.96 second waveform -> Mel spectrogram -> log scale -> 96 (frames) × 64 (Mel bins)
- The VGGish model is a 2D CNN
- Processes the log Mel spectrogram as 2D image.
- **Output:** 128-dimensional feature vector set

Pose

**Xie, Y.F., Zhang, S., Liu, Y.D. (2021).
Abnormal behavior recognition in
classroom pose estimation of college
students based on spatiotemporal
representation learning**

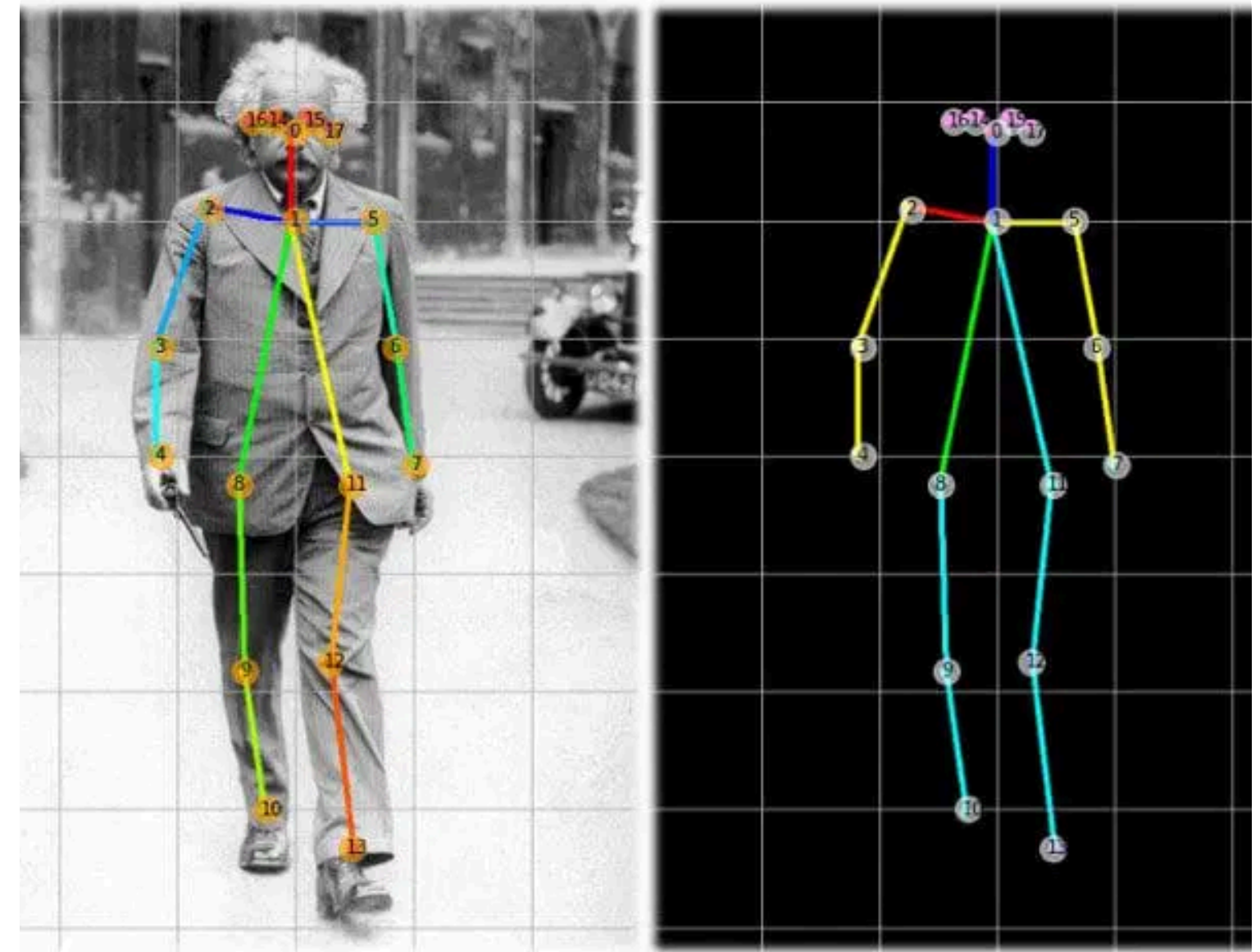
- achieved 96% AP on Student-Class-Behavior-Dataset



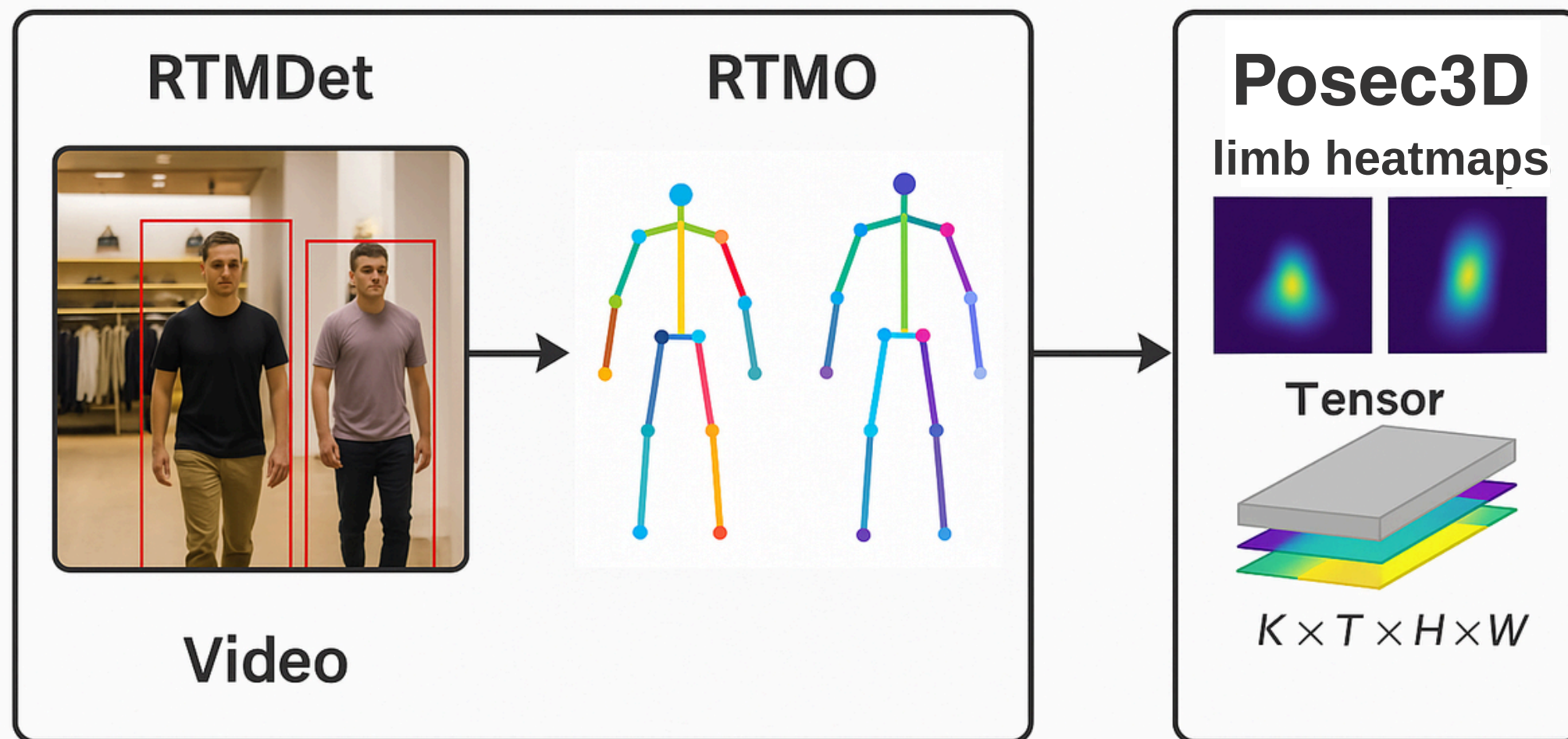
Pose Feature Extraction

Advantages

- Clear semantic information
- Stability against noise factors
- Detect abnormal behaviors with weak motion
- Combine temporal information to differentiate behaviors

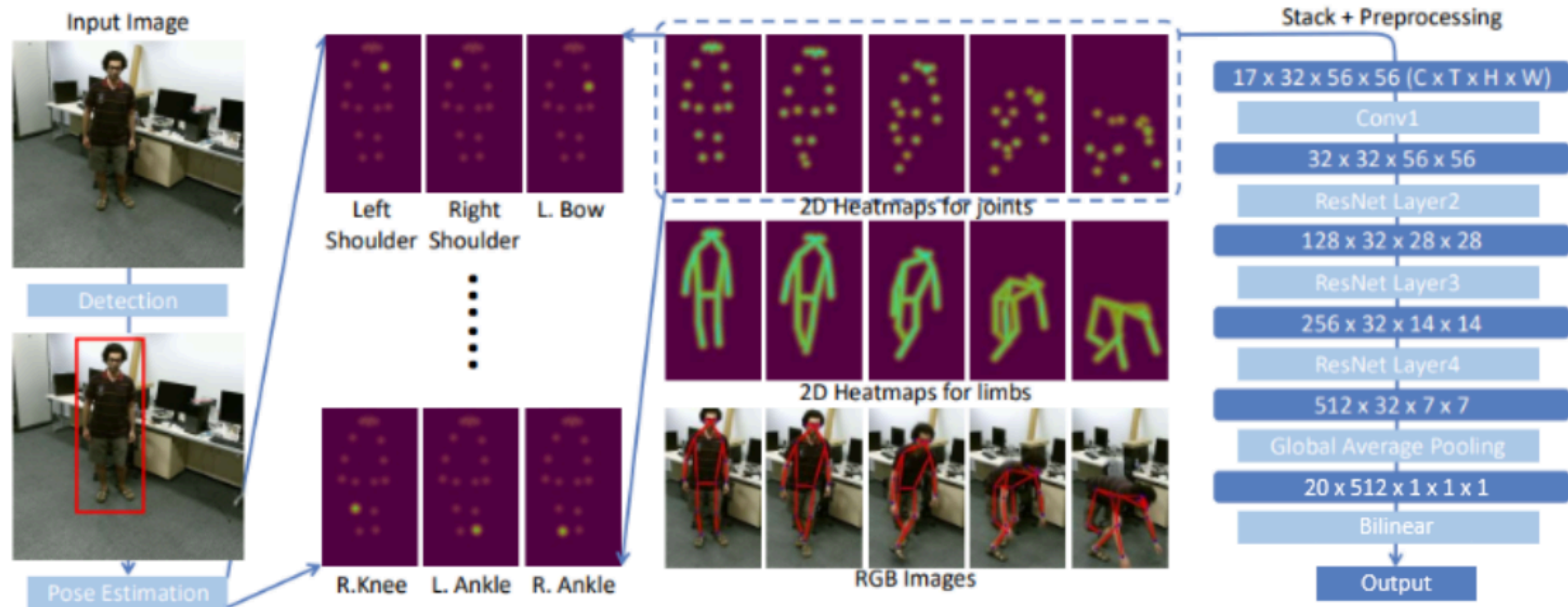


Pose feature extraction



- Balance of Speed, Accuracy
- Maintains Contextual Consistency across frames

Pose feature extraction



Posec3D

MFMS Alignment Module

Projection Layer:

- Converts each stream to a common 256-dimensional space
- Ensures shape compatibility for interaction

Self-Attention:

- Focuses on informative frames within each modality
- Filters out low-signal or redundant inputs

Cross-Attention:

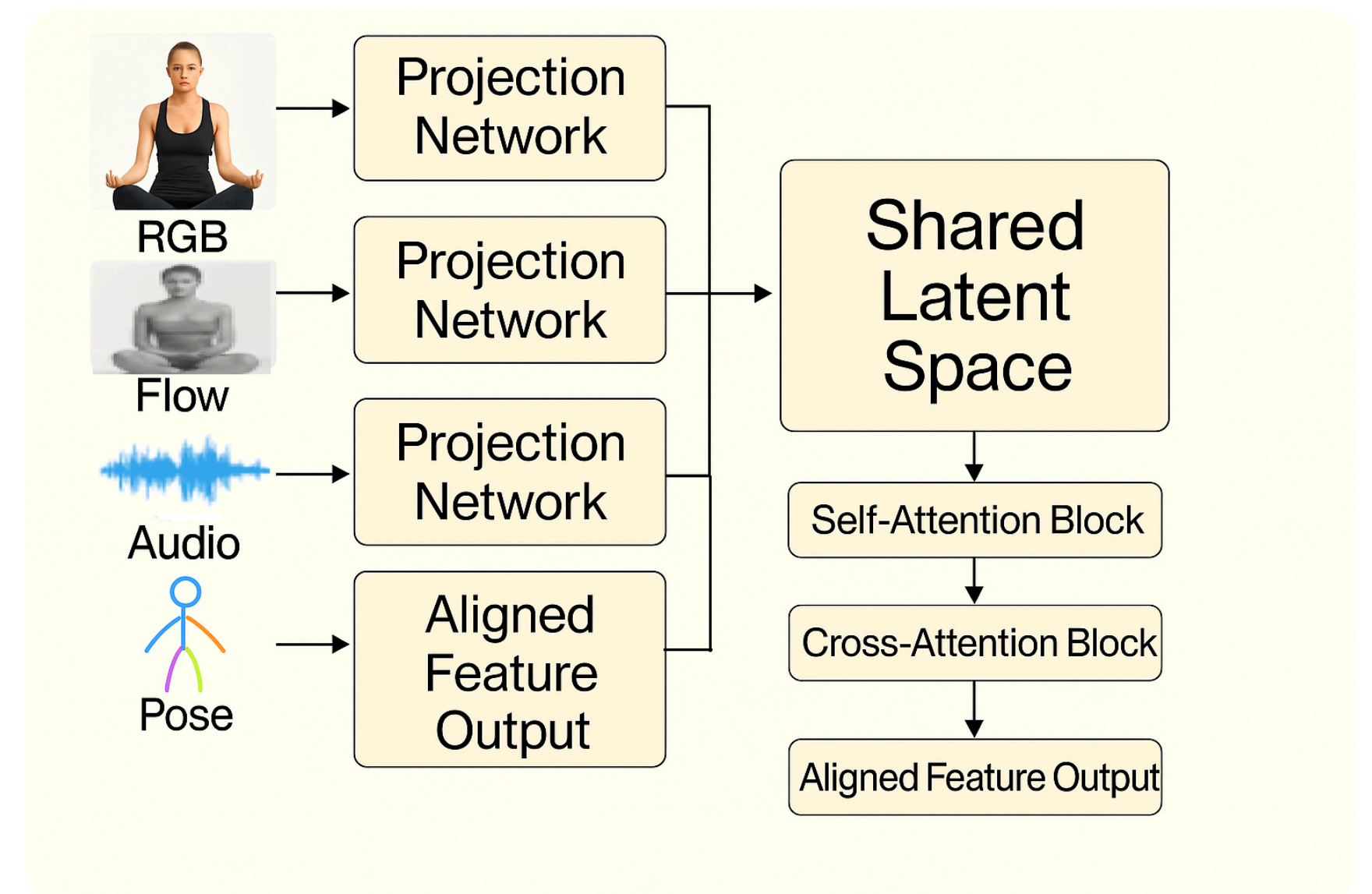
- Enables information exchange across modalities
- Boosts weak signals with complementary ones

Outcome

- Aligns features in both time and meaning
- Prepares unified features for better fusion

Why Alignment?

- Modalities differ in semantics, time scale, and feature distribution
- Direct fusion leads to mismatched and misaligned features



Gated Cross-Modal Attention for Fusion

Why Gated Fusion?

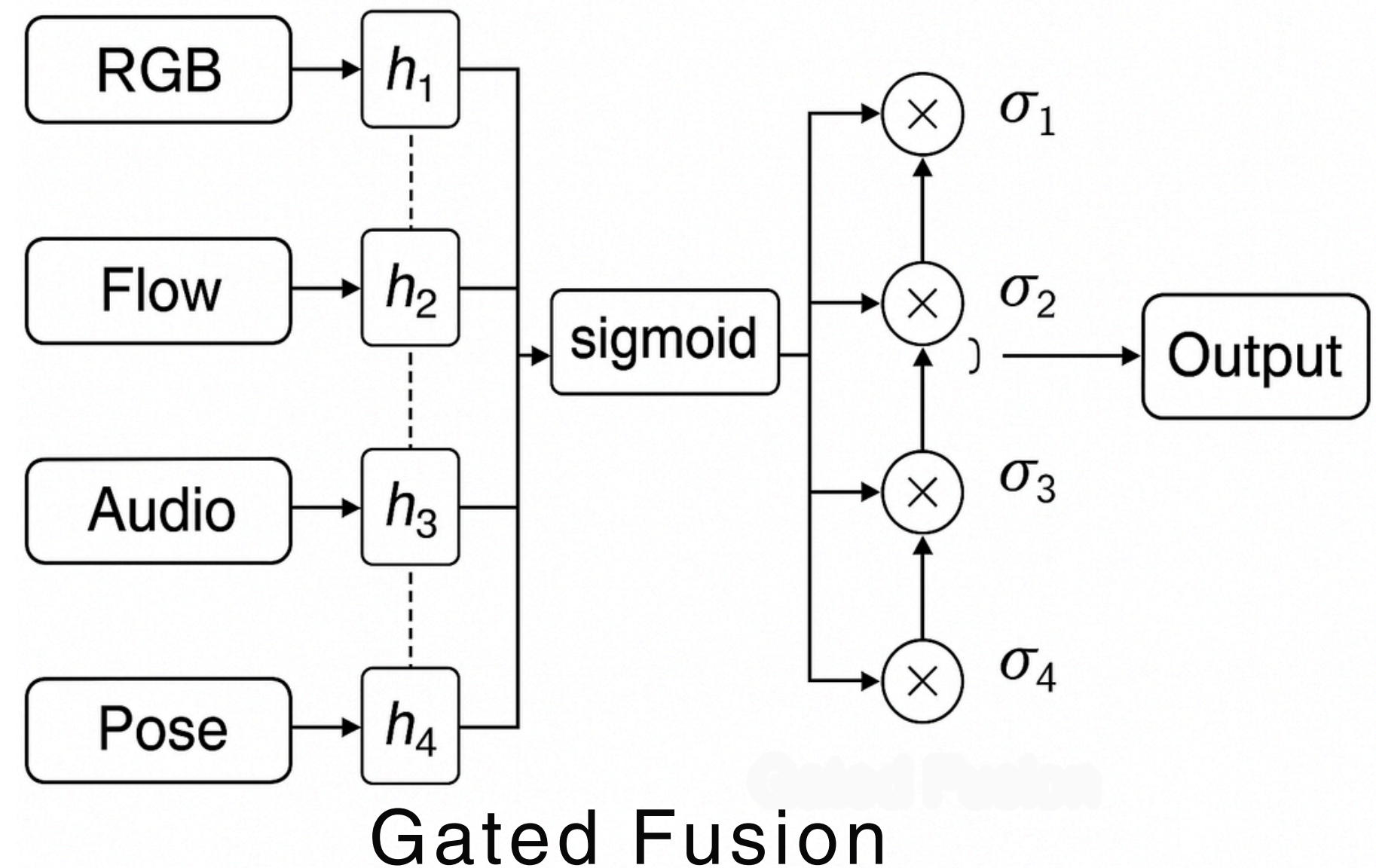
- Not all modalities are equally reliable at all times
- Direct fusion risks overtrusting noisy inputs

Fusion Mechanism:

- Project each modality to the same dimension
- Compute importance score for each modality (via similarity + sigmoid)
- Use soft gating mask $\in [0, 1]$ to control each stream's contribution
- Final fused feature = weighted sum across all gated streams

Benefits

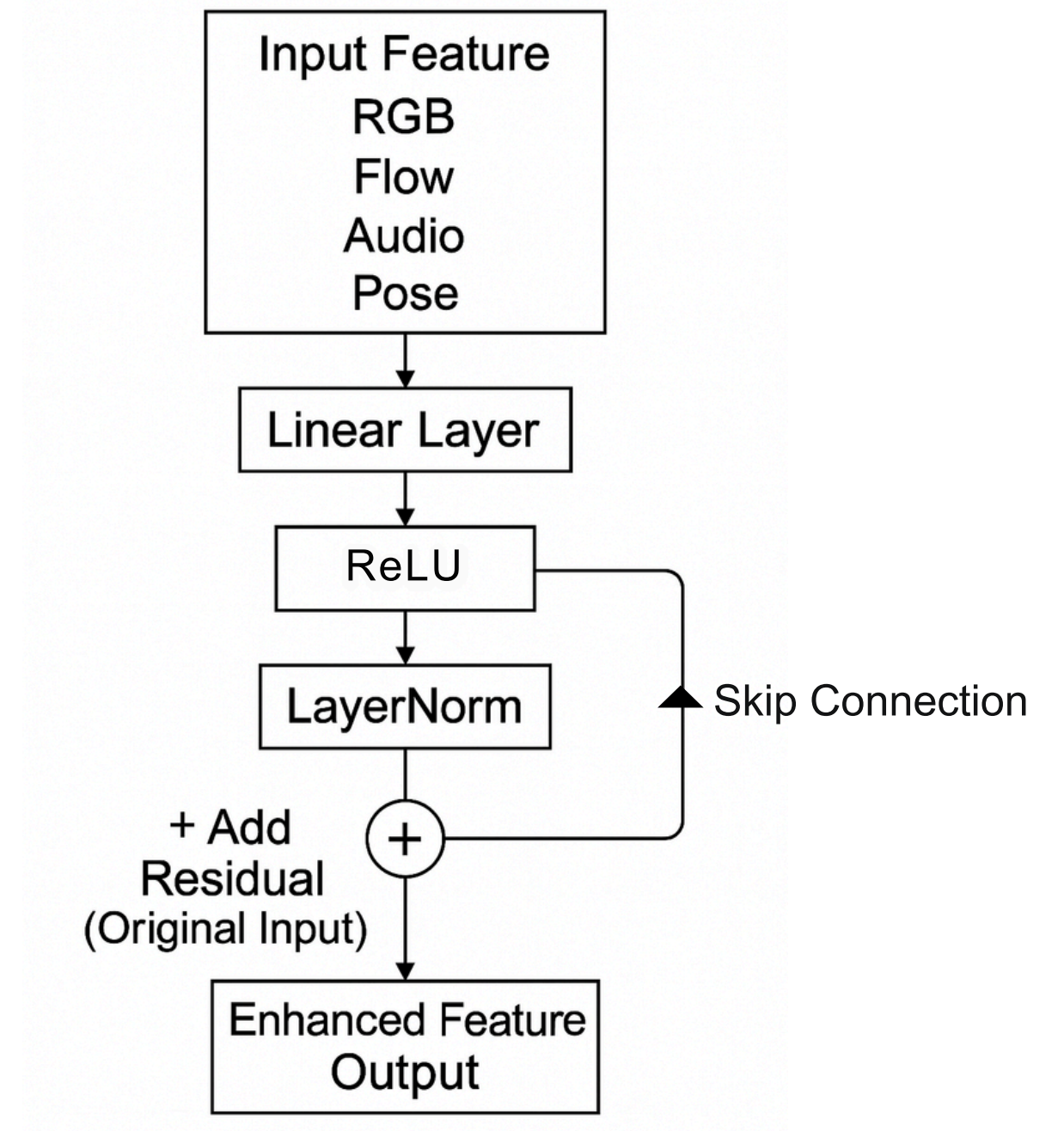
- Selectively focuses on trustworthy modalities
- Robust against noise and missing signals



Preserve Weak Modalities with Residual Connection

Why Residual Connection?

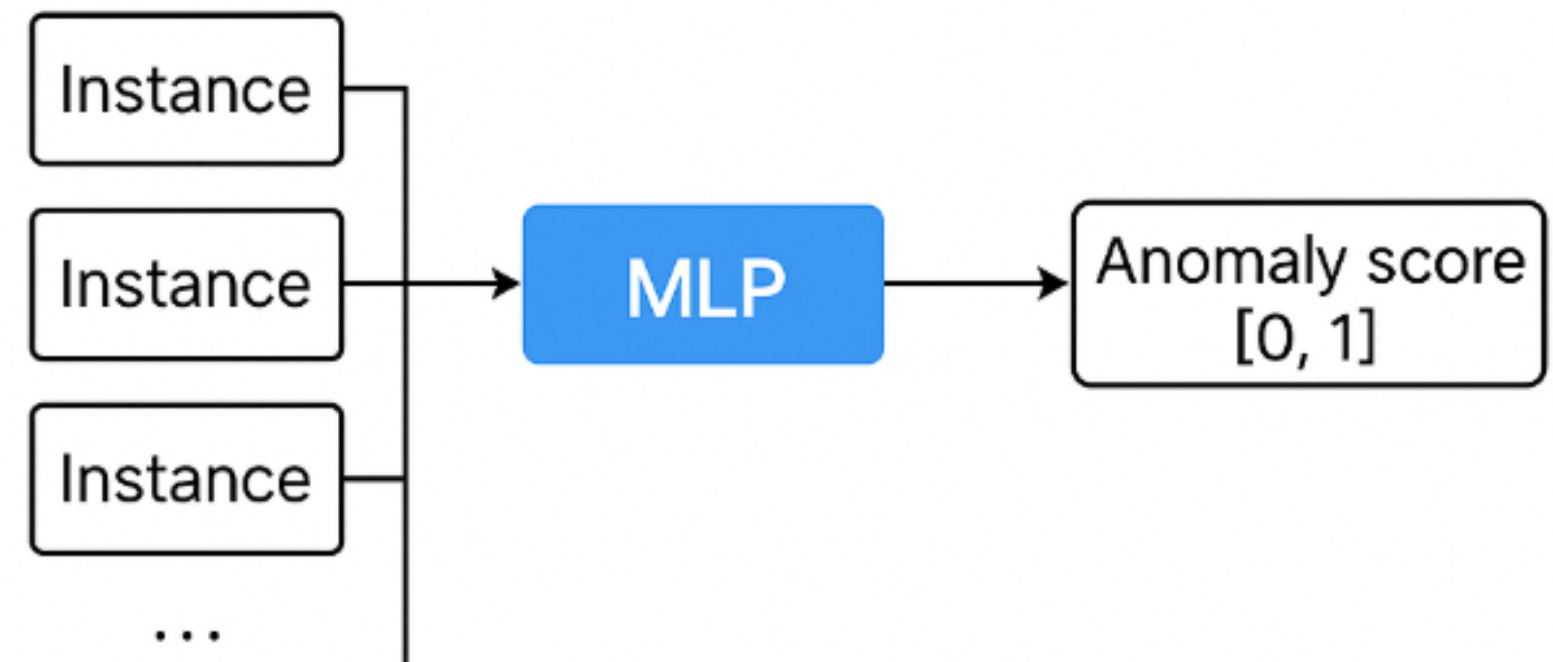
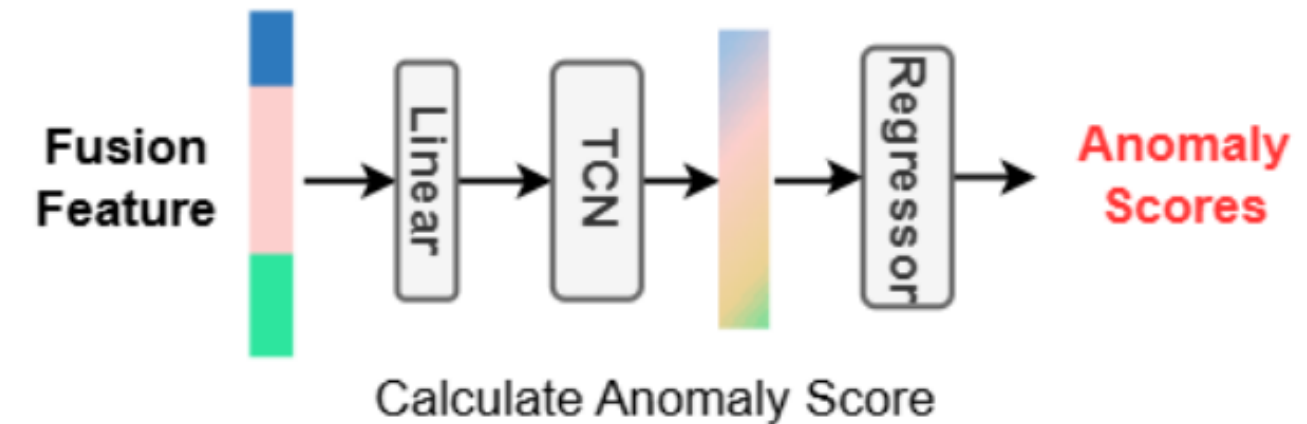
- Fusion may suppress Pose or Audio signals
- Apply Linear layer and LayerNorm to fusion output
- Add back the original feature (residual connection)
- Maintain both fused and original information
- Avoid bias toward dominant modalities like RGB or Flow



Anomaly Scoring

Anomaly Scoring via MIL

- TCN captures temporal patterns across fused features.
- Each frame is assigned an anomaly score by a MIL regressor.
- Top-k scoring frames are selected and averaged.
- Final score in $[0, 1]$ indicates video-level abnormality.



Structured Loss & Why Contrastive?

Loss Components:

Unimodal MIL Loss (UMIL):

- Learn strong features per modality.

Modality Alignment Loss (MAL):

- Align modalities in shared space.

Multimodal MIL Loss (MMIL):

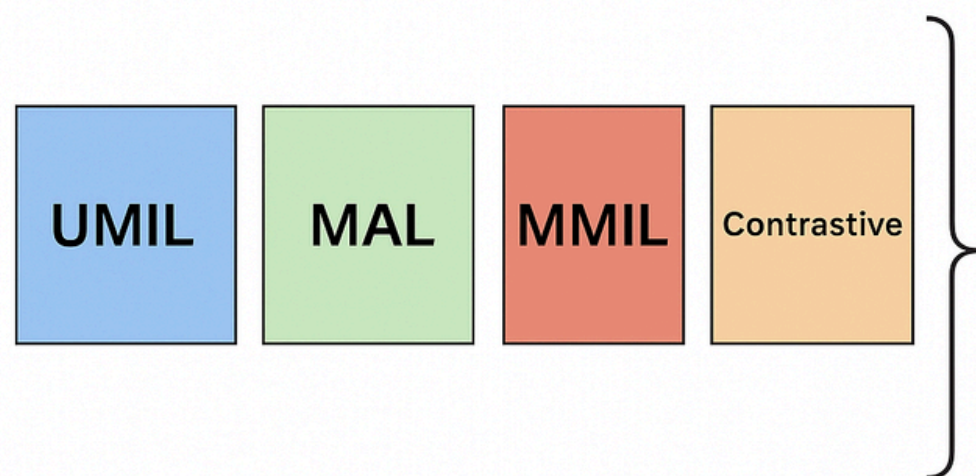
- Detect anomalies from fused features.

Contrastive Loss:

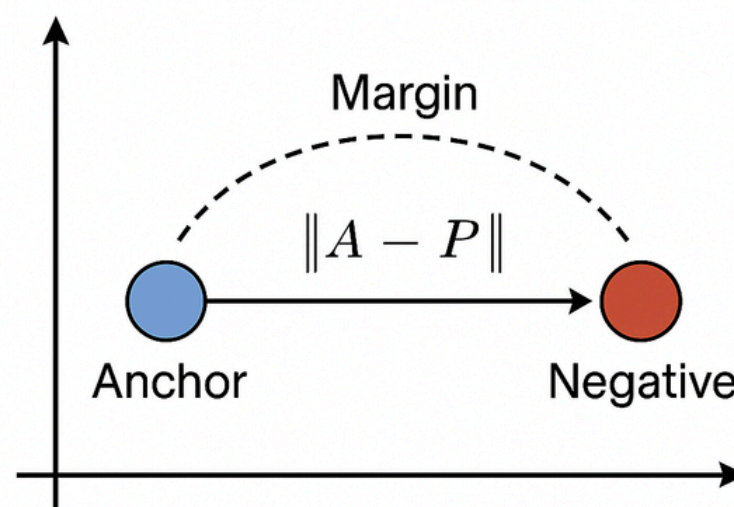
- Boost feature separability after alignment.

Why Contrastive Loss?

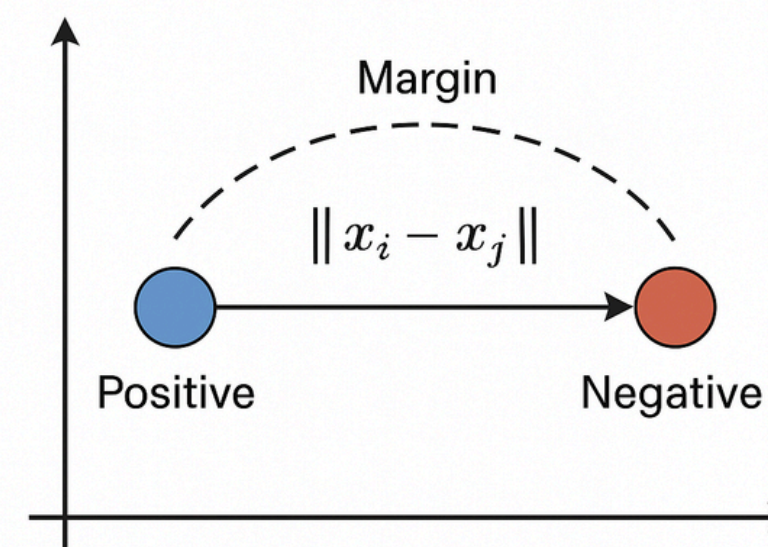
- Easier than Triplet under weak labels
- Needs only similar/dissimilar frame pairs
- Positive: frames from same abnormal video
- Negative: frames from normal vs. abnormal video



Triplet Loss



Contrastive Loss



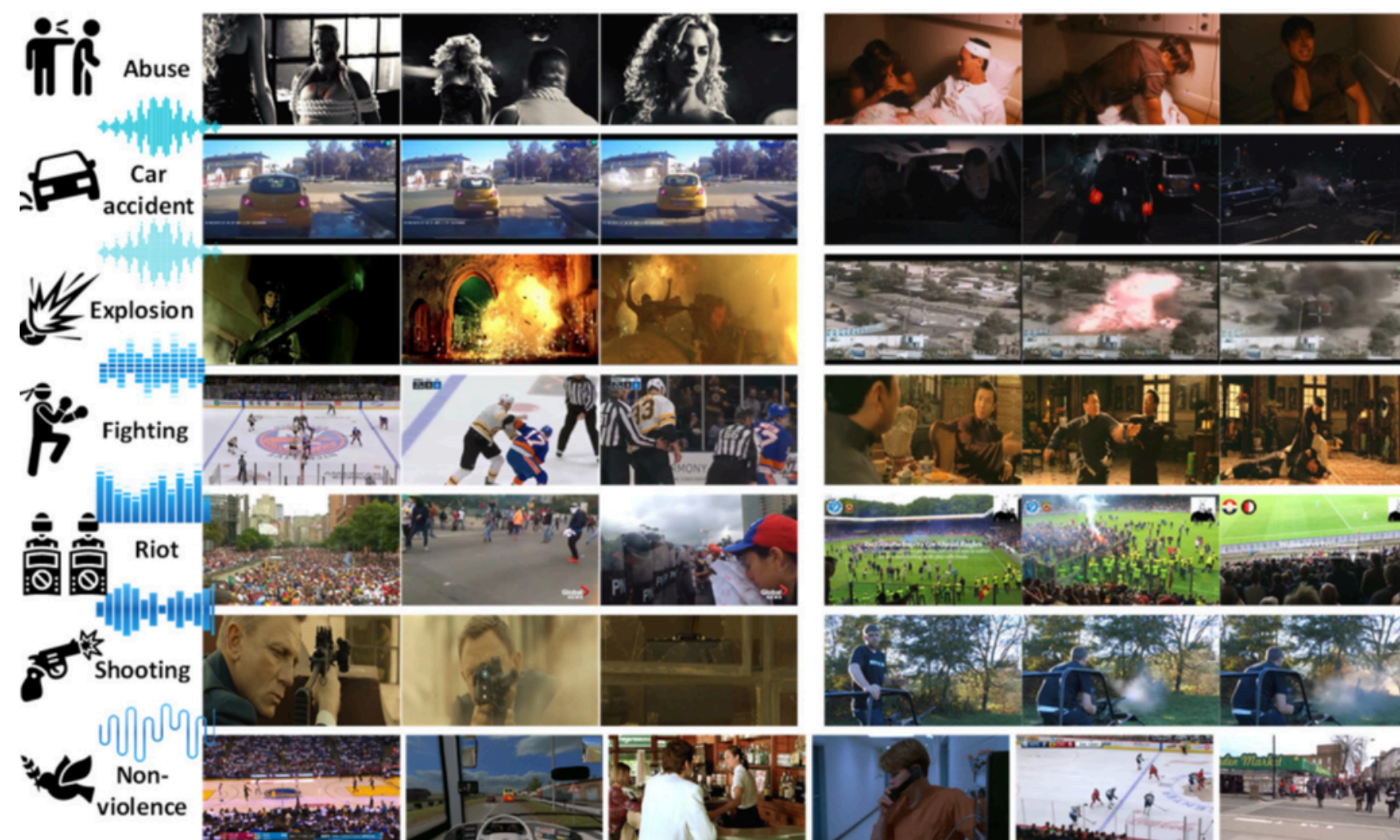
Experimental

OVERVIEW XD-VIOLENCE DATASET

~4,000 videos ($\approx$ 217 hours)

Movies, vlogs, sports broadcasts, news footage

Abuse, Car Accident, Explosion, Fighting, Riot, Shooting, Non-violence



XD-Violence Dataset

Experimental Dataset and Evaluation Metrics

Why It Fits Our Study

- Real-world Complexity
- Weak Supervision
- Multimodal Learning
- Benchmark & reproducibility



Person fighting



Gun violence



Explosion



Stabbing

Experimental

Dataset and Evaluation Metrics

Evaluation Metrics

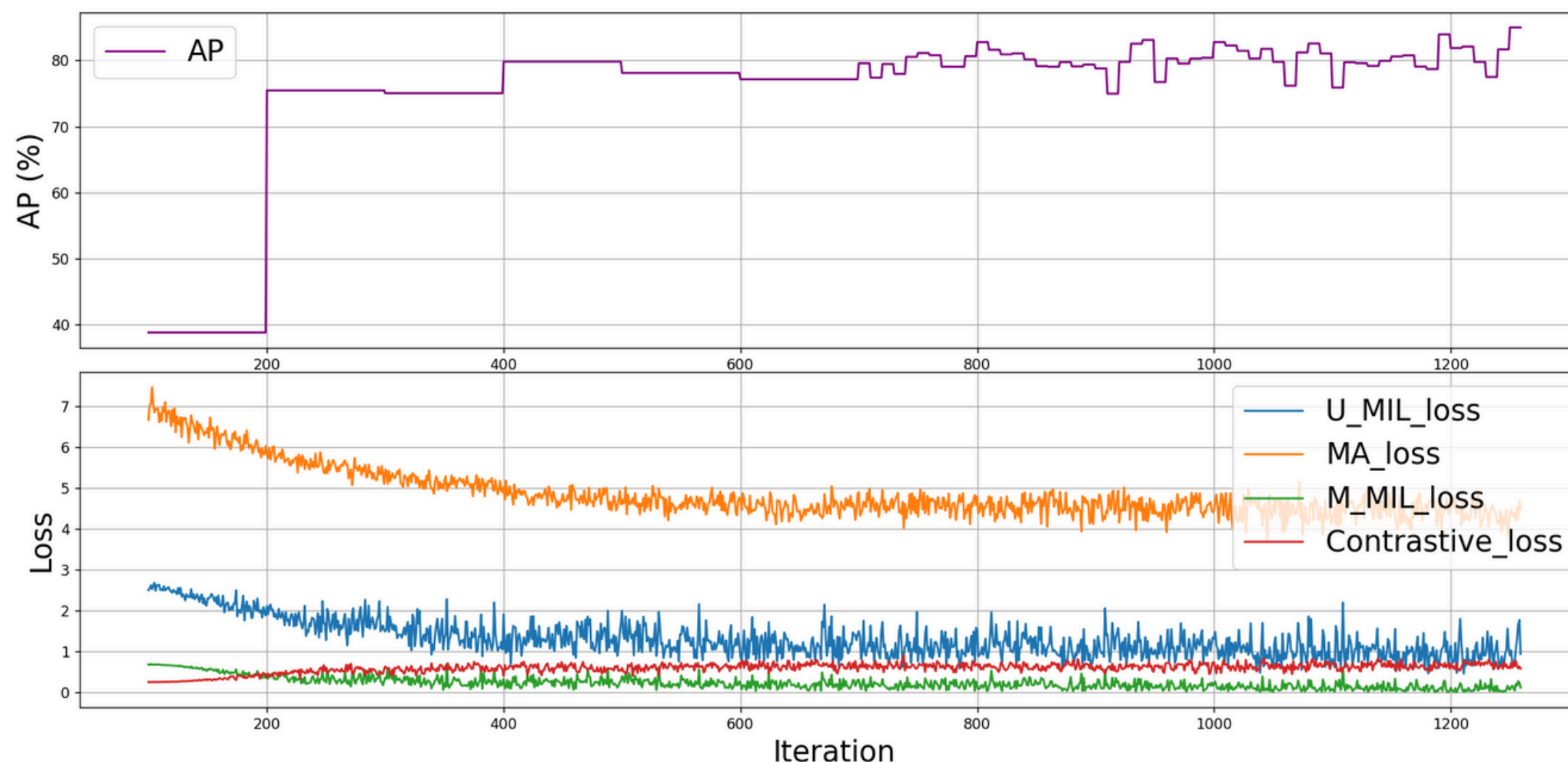
mAP: 84.98%

U-MIL Loss: 0.925

MA Loss: 4.3241

M-MIL Loss: 0.0341

Contrastive Loss: 0.6939

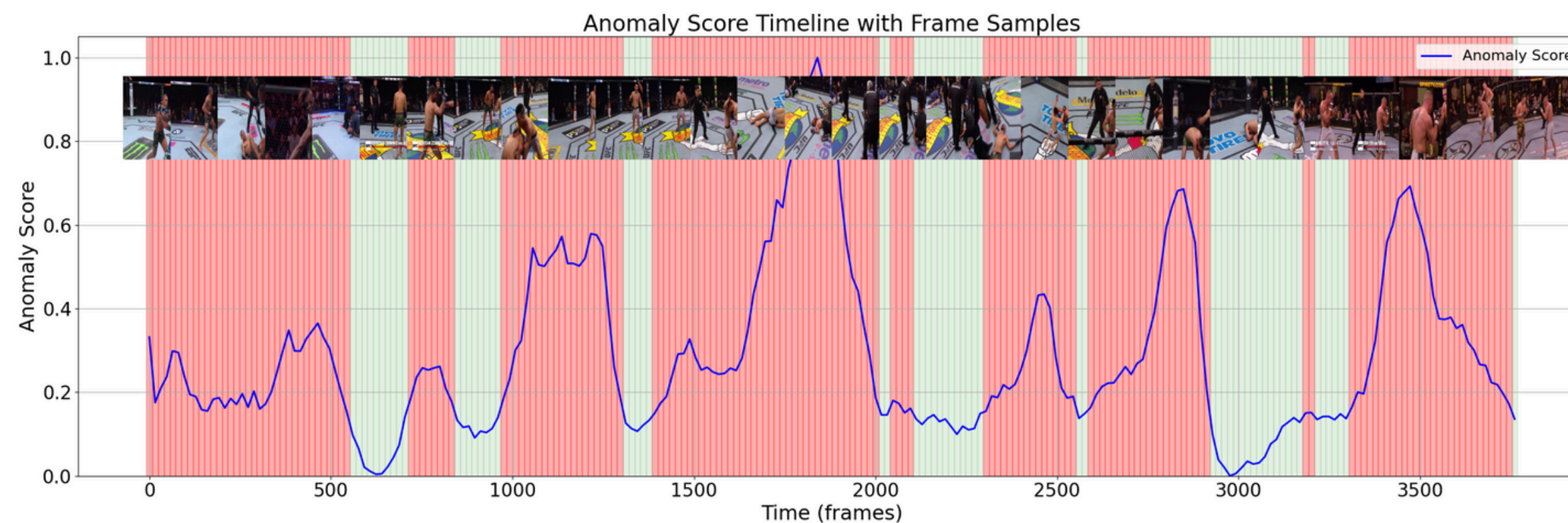


Frame-level Average Precision (AP)

Experimental

Generalization on Out-of-Distribution Videos

Achieved 46.09 F1-score, accurately detecting violent actions in unseen UFC fight video



Area Under the Curve (AUC)

Experimental

Comparisons with Other Methods

Method	Modality	AP (%)
SVM baseline	RGB	50.78
OCSVM [28]	RGB	27.25
Hasan et al. [14]	RGB	30.77
Sultani et al. [30]	RGB	75.68
RTFM [34]	RGB	77.81
HL-Net [39]	RGB + Audio	78.64
ACF [37]	RGB + Audio	80.13
Zhang et al. [41]	RGB + Audio	81.43
MSBT [31]	RGB + Audio + Flow	84.32
HyperVD [26]	RGB + Audio	85.67
MAVD [16]	RGB + Audio + Flow	86.07
CFA-HLGAtt [13]	RGB + Audio	86.34
Ours	RGB + Audio + Flow	83.51
Ours	RGB + Audio + Flow + Pose	84.98

Conclusion

Problem: Weakly supervised anomaly detection in surveillance videos.

Goal: Improve detection precision.

Proposed Solution:

- Add pose stream to original RGB, Flow, and Audio streams.
 - Use MFMS for semantic-level feature alignment.
 - Apply GMF for context-aware multimodal fusion.
 - Replace Triplet Loss with Contrastive Loss to enhance feature discrimination.
-

Conclusion

Performance on XD-Violence dataset:

- Achieved 84.98% Average Precision, surpassing strong baselines.
- Ablation Studies confirm the impact of each component:
 - Pose stream: +0.83%
 - GMF: +0.64%
 - Contrastive Loss: +0.64%
 - MFMS: +0.78%

Accepted paper at EvoStar 2025

Conclusion

Current Challenges:

- Training plateaus after ~850 epochs.
- Sensitive to noisy modalities (audio, crowded pose estimations).
- Decreased performance on anomalies without clear human activity.
- Limited by hardware constraints

Future Work:

- Add data diversity and accurate labeling
 - Deeper ablation to separately evaluate the contributions of MFMS and GMF.
 - Integrate dynamic loss reweighting strategies to improve training efficiency.
 - Investigate real-time optimization for practical surveillance deployment.
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Question & Answer

Thank You
